

Exercícios - Capítulo 03

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1. Encontre todos os voos que:
 - a. Tiveram um atraso de duas horas ou mais na chegada.

```
library(nycflights13)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
filter(flights, arr_delay >= 120)
```

A tibble: 10,200 x 19

	year	month	day	dep_time	sched_dep_time	dep_delay	arr_time	sched_arr_time
	<int>	<int>	<int>	<int>	<int>	<dbl>	<int>	<int>
1	2013	1	1	811	630	101	1047	830
2	2013	1	1	848	1835	853	1001	1950
3	2013	1	1	957	733	144	1056	853
4	2013	1	1	1114	900	134	1447	1222
5	2013	1	1	1505	1310	115	1638	1431

```

6 2013      1      1      1525          1340          105          1831          1626
7 2013      1      1      1549          1445           64          1912          1656
8 2013      1      1      1558          1359          119          1718          1515
9 2013      1      1      1732          1630           62          2028          1825
10 2013     1      1      1803          1620          103          2008          1750

```

```
# i 10,190 more rows
```

```

# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dtm>

```

b. Foram para Houston (IAH ou HOU).

```
filter(flights, dest %in% c("IAH", "HOU"))
```

```
# A tibble: 9,313 x 19
```

```

  year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
  <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
1  2013     1     1     517           515           2     830           819
2  2013     1     1     533           529           4     850           830
3  2013     1     1     623           627          -4     933           932
4  2013     1     1     728           732          -4    1041          1038
5  2013     1     1     739           739           0    1104          1038
6  2013     1     1     908           908           0    1228          1219
7  2013     1     1    1028          1026           2    1350          1339
8  2013     1     1    1044          1045          -1    1352          1351
9  2013     1     1    1114           900        134    1447          1222
10 2013     1     1    1205          1200           5    1503          1505

```

```
# i 9,303 more rows
```

```

# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dtm>

```

c. Foram operados pela United, American ou Delta.

```
filter(flights, carrier %in% c("UA", "AA", "DL"))
```

```
# A tibble: 139,504 x 19
```

```

  year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
  <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
1  2013     1     1     517           515           2     830           819
2  2013     1     1     533           529           4     850           830
3  2013     1     1     542           540           2     923           850

```

```

4 2013      1      1      554           600          -6      812           837
5 2013      1      1      554           558          -4      740           728
6 2013      1      1      558           600          -2      753           745
7 2013      1      1      558           600          -2      924           917
8 2013      1      1      558           600          -2      923           937
9 2013      1      1      559           600          -1      941           910
10 2013     1      1      559           600          -1      854           902
# i 139,494 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dtm>

```

d. Partiram em julho, agosto e setembro.

```
filter(flights, month %in% seq(7,9))
```

```

# A tibble: 86,326 x 19
   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
1  2013     7     1       1          2029          212     236          2359
2  2013     7     1       2          2359           3     344           344
3  2013     7     1      29          2245         104     151           1
4  2013     7     1      43          2130         193     322           14
5  2013     7     1      44          2150         174     300           100
6  2013     7     1      46          2051         235     304          2358
7  2013     7     1      48          2001         287     308          2305
8  2013     7     1      58          2155         183     335           43
9  2013     7     1     100          2146         194     327           30
10 2013     7     1     100          2245         135     337          135
# i 86,316 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dtm>

```

e. Chegaram com mais de duas horas de atraso, mas não saíram atrasados.

```
filter(flights, arr_delay > 120, dep_delay <= 0)
```

```

# A tibble: 29 x 19
   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
   <int> <int> <int>   <int>         <int>         <dbl>   <int>         <int>
1  2013     1    27     1419          1420          -1     1754          1550

```

```

2 2013 10 7 1350 1350 0 1736 1526
3 2013 10 7 1357 1359 -2 1858 1654
4 2013 10 16 657 700 -3 1258 1056
5 2013 11 1 658 700 -2 1329 1015
6 2013 3 18 1844 1847 -3 39 2219
7 2013 4 17 1635 1640 -5 2049 1845
8 2013 4 18 558 600 -2 1149 850
9 2013 4 18 655 700 -5 1213 950
10 2013 5 22 1827 1830 -3 2217 2010
# i 19 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dtm>

```

f. Atrasaram pelo menos uma hora, mas compensaram mais de 30 minutos durante o trajeto.

```
filter(flights, dep_delay >= 60, (dep_delay - arr_delay) > 30)
```

```

# A tibble: 1,844 x 19
  year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
  <int> <int> <int>   <int>         <int>      <dbl>    <int>         <int>
1 2013     1     1    2205         1720        285      46         2040
2 2013     1     1    2326         2130        116     131          18
3 2013     1     3    1503         1221        162    1803        1555
4 2013     1     3    1839         1700         99    2056        1950
5 2013     1     3    1850         1745         65    2148        2120
6 2013     1     3    1941         1759        102    2246        2139
7 2013     1     3    1950         1845         65    2228        2227
8 2013     1     3    2015         1915         60    2135        2111
9 2013     1     3    2257         2000        177      45        2224
10 2013     1     4    1917         1700        137    2135        1950
# i 1,834 more rows
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dtm>

```

g. Saíram entre meia-noite e 6h (incluindo esses horários)

```
filter(flights, dep_time %in% seq(0, 600))
```

```

# A tibble: 9,344 x 19
  year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
  <int> <int> <int>   <int>         <int>      <dbl>    <int>         <int>

```

	<int>	<int>	<int>	<int>	<int>	<dbl>	<int>	<int>
1	2013	1	1	517	515	2	830	819
2	2013	1	1	533	529	4	850	830
3	2013	1	1	542	540	2	923	850
4	2013	1	1	544	545	-1	1004	1022
5	2013	1	1	554	600	-6	812	837
6	2013	1	1	554	558	-4	740	728
7	2013	1	1	555	600	-5	913	854
8	2013	1	1	557	600	-3	709	723
9	2013	1	1	557	600	-3	838	846
10	2013	1	1	558	600	-2	753	745

i 9,334 more rows

i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
hour <dbl>, minute <dbl>, time_hour <dtm>